Define Problem Statements

Wandr AI – AI Travel Itinerary Planner

1. Problem Statement (HMW Format)

How Might We help travellers — from solo backpackers to family vacation planners — quickly generate a personalised, realistic, and budget-aware day-by-day travel itinerary without spending hours researching multiple sources, so that they can focus on enjoying the experience rather than planning logistics?

2. Root Cause Analysis

Pain Point	Root Cause	Current Workaround
Takes hours to plan a multi-day trip	No single tool combines all planning dimensions	Manual research across 5-10 websites
Budget overruns are common	Costs are not tracked during planning	Spreadsheet after the fact
Plans don't adapt to traveller style	Generic recommendations ignore preferences	Ignore recommendations
Changing one day breaks the whole plan	No tool supports partial regeneration	Redo the entire plan manually

3. Problem Scope & Constraints

- **In scope:** Destination, date range, number of travellers, budget, travel style, pace, interests, dietary/accessibility needs.
- **Out of scope (v1):** Real-time pricing, live hotel booking, flight search, group trip collaboration.
- **Technical constraint:** AI generation must complete within 30 seconds (Groq timeout set to 120 s).
- **Security constraint:** No user can view another user's trips (enforced by Supabase RLS + JWT verification).

4. Success Criteria

Criterion	Target	Measurement
AI generation time	< 30 seconds	Logged per request
User authentication success rate	> 99%	Supabase Auth logs
Generated itinerary relevance	User rates ≥ 4/5	Post-trip survey (future)
Budget accuracy	Estimated within ±20%	User feedback
CORS security	Zero cross-origin leaks	Pytest CORS suite – 9/9 pass